Synthesis of 4-Halo-2(5*H*)-furanones and Their Suzuki-Coupling Reactions with Organoboronic Acids. A General Route to 4-Aryl-2(5*H*)-furanones†

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4-Halo-2(5*H*)-furanones were prepared by the halolactonization of 2,3-allenoic acids. The subsequent Suzuki coupling reaction of 4-halo-2(5*H*)-furanones with aryl boronic acids was carried out to produce 4-aryl-2(5*H*)-furanones in excellent yields.

Keywords 2,3-Allenoic acid, halolactonization, Suzuki coupling, 4-halo-2(5*H*)-furanone, 4-aryl-2(5*H*)-furanone

Introduction

Recently much attention has been focused on the synthesis of polysubstituted 2(5*H*)-furanones due to their potential biological activities, which might be used as potential insecticides, bactericides, fungicides, antibiotics, anticancer agents, anti-inflammatories, allergy inhibitors, antisoriasis agents, cycloxygenase inhibitors, and phospholipase A_2 inhibitors, *etc*.[1] Due to the substituent-loading capability, acidity, and the chirality of 2,3-allenoic acids, we developed some transition metal-catalyzed or mediated methods for the efficient synthesis of 2(5*H*)-furanones starting from 2,3-allenoic acids[2-4] and 2,3-allenoates.[5] Furthermore, we developed a high-yielding procedure for the synthesis of β-halobutenolides and the subsequent coupling reactions with terminal alkynes or organozinc reagents to afford β-differently substituted butenolides.[6,7] In this paper, the scope of this halolactonization of polysubstituted 2,3-allenoic acids and the corresponding subsequent Pd(0)-catalyzed Suzuki coupling reaction with aryl/alkenyl boronic acids will be described.

Results and discussion

Iodolactonization of polysubstituted 2,3-allenoic acids

According to a step-wise procedure of treatment of 2,3-allenoic acids with I_2 and K_2CO_3,[6,7] polysubstituted 2,3-allenoic acids can be iodolactonized under the mild conditions to afford β-iodobutenolides in high yields. The results were summarized in Table 1. In Table 1, the following points should be noted: (1) the steric hinderance of the substituents has very limited influence on the yields of iodolactonization (Entries 1—3, Table 1); (2) the reaction could be carried out in high yields for mono or polysubstitued differently structured 2,3-allenoic acids.

Bromolactonization of polysubstituted 2,3-allenoic acids

The β-bromobutenolides can be prepared by the treatment of 2,3-allenoic acids with NBS and K_2CO_3 in water,[6,7] and the results were summarized in Table 2.

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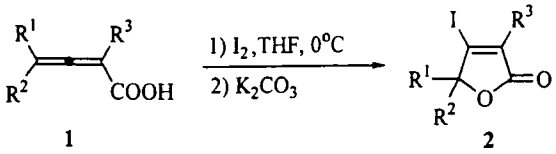
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†Dedicated to Professor HUANG Wei-Yuan on the occasion of his 80th birthday.

From the data in Table 2, it can be concluded that thedifferently substituted 2,3-allenoic acids afforded β-bromobutenolides in good to excellent yields.

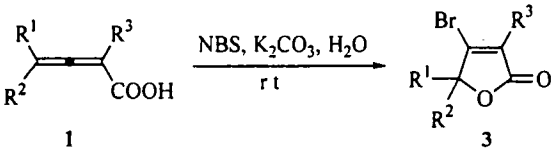
Table 1 Iodolactonization reaction of 2,3-allenoic acids



1 2

Entry	R^1	R^2	R^3	1	2	Yield (%)
1	*n*-Bu	H	H	1a	2a	91[6,7]
2	*n*-C_7H_{15}	H	H	1b	2b	90[6,7]
3	*t*-Bu	H	H	1c	2c	91
4	Ph	H	Me	1d	2d	99
5	Ph	H	*n*-Pr	1e	2e	97
6	1′-Naphthyl	H	Me	1f	2f	97
7	Ph	Me	H	1g	2g	87
8	*p*-BrC_6H_4	H	*n*-Pr	1h	2h	96

Table 2 Bromolactonization reaction of 2,3-allenoic acids[a]



1 3

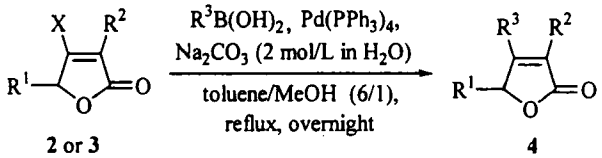
Entry	R^1	R^2	R^3	1	3	Yield (%)[b]
1	*n*-Bu	H	H	1a	3a	84[6,7]
2	*n*-C_7H_{15}	H	H	1b	3b	81[6,7]
3	Ph	H	Me	1d	3c	94
4	Ph	H	*n*-Pr	1e	3d	99
5	1′-Naphthyl	H	Me	1f	3e	98
6	Ph	Me	H	1g	3f	99

[a] 1/NBS/K_2CO_3 = 1:1.1:0.5; [b] isolated yield.

Suzuki coupling reaction of 4-halo-2(5H)-furanones with aryl boronic acids

The Suzuki coupling reaction of an organic halide with aryl/alkenyl boronic acids provides one of the most efficient methods for the formation of C—C bonds.[8] We examined the behavior of differently structured 4-halo-2(5*H*)-furanones and organoboronic acids, the results were summarized in Table 3. From Table 3, it can be concluded that the Suzuki coupling of 4-halo-2(5*H*)-furanones with aryl boronic acids afforded the corresponding 4-aryl-2(5*H*)-furanones in high yields. Both iodides and bromides worked with similar yields (compare entry 5 with 6 and entry 8 with 9, Table 3). However, the coupling reaction did not occur when methylboronic acid was used (Entry 10, Table 3).

Table 3 Suzuki coupling reaction of 4-halo-2(5*H*)-furanones with aryl/alkenyl boronic acids[a]



2 or 3 4

Entry	R^1	R^2	X	2 or 3	R^3	Product	Yield (%)[b]
1	*n*-Bu	H	I	2a	Ph	4a	93
2	*n*-Bu	H	I	2a	*o*-$MeOC_6H_4$	4b	88
3	*n*-C_7H_{15}	H	I	2b	Ph	4c	88
4[c]	Ph	Me	I	2d	Ph	4d	84
5	Ph	Me	I	2d	Ph	4d	99
6	Ph	Me	Br	3c	Ph	4d	97
7	Ph	Me	I	2d	*o*-$MeOC_6H_4$	4e	85
8	1′-Naphthyl	Me	I	2f	Ph	4f	87
9	1′-Naphthyl	Me	Br	3e	Ph	4f	83
10	*n*-Bu	H	I	2a	Me	–	–

[a] 2 or 3/$R^3B(OH)_2$/$Pd(PPh_3)_4$ = 1/2/0.05; [b] isolated yield; [c] xylenes were used instead of toluene as solvent.

Conclusion

In conclusion, we prepared β-iodo (or bromo) butenolides *via* the iodo/bromolactonization of polysubstituted 2,3-allenoic acids. Their Pd(0)-catalyzed Suzuki coupling reaction with organoboronic acids provided a simple and efficient route to 4-aryl/alkenyl substituted 2(5*H*)-furanones.

Experimental

Apparatus

^1H NMR spectra were obtained on a 300 MHz Varian Spectrometer with TMS as the internal standard. Mass spectra were obtained by using an HP5989A Spectrometer at an ionization potential of 70 eV. IR spectra were

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recorded by using a Perkin-Elmer Lambda 2 spectrometer. Melting points were not corrected.

Reagents and substrates

2,3-Octadienoic acid, 2,3-undecadienoic acid and 5,5-dimethyl-2,3-hexadienoic acid were prepared according to a published procedure[9] *via* the reaction of CO_2 with the corresponding 1,2-allenic lithiums, which, in turn, were prepared from the treatment of the corresponding 1,2-allenes with *n*-BuLi. 4-Phenyl-3-methyl-2,3-butadienoic acid was prepared according to the known method[10] by the treatment of acyl chlorides with ethyl 2-(triphenylphoranylidene) propionates and followed by hydrolysis of the *in-situ* formed 2,3-allenoates with 1.5 equiv NaOH or LiOH. Iodine, NBS and aryl boronic acids were commercially available and used as it is.

General procedure for the preparation of 4-*iodo*-5-(*t*-*butyl*)-2(5*H*)-*furanone* (**2c**)

A solution of 5,5-dimethyl-2,3-hexadienoic acid (**1c**) (28.0 mg, 0.20 mmol) and iodine (61.0 mg, 0.24 mmol) in 2 mL of THF was stirred at 0℃ for 1 h. Then K_2CO_3 (27.6 mg, 0.2 mmol) was added quickly at 0℃ and stirred for 6 h at this temperature. The reaction mixture was quenched with an aqueous sodium thiosulfate solution, extracted with diethyl ether (3 × 10 mL), and dried over anhydrous $MgSO_4$. After removal of the solvent, the residue was purified by flash chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 10/1) to afford 48.6 mg (91%) of 4-iodo-5-(*t*-butyl)-2(5*H*)-furanone (**2c**). Yellow solid, m.p. 78—79℃ (*n*-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 6.63 (d, J = 1.5 Hz, 1H), 4.74 (d, J = 1.5 Hz, 1H), 1.12 (s, 9H); IR (KBr) ν: 1768, 1646, 1170 cm^{-1}; MS (EI) *m/z* (%): 267 (M^+ + 1, 3.98), 57 (100); Anal. calcd for $C_8H_{11}IO_2$: C 36.11, H 4.17; found C 36.18, H 3.89.

The data for compounds **2a** and **2b** were reported in Ref. 7.

3-*Methyl*-4-*iodo*-5-*phenyl*-2(5*H*)-*furanone* (**2d**) Starting from 470.0 mg (2.70 mmol) of **1d**, 720.0 mg (2.84 mmol) of iodine, and 414.6 mg (2.97 mmol) of K_2CO_3 to afford 800.0 mg (99%) of **2d**. Yellow solid, m.p. 93—94℃ (*n*-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 7.48—7.38 (m, 3H), 7.30—7.18 (m, 2H), 5.69 (q, J = 1.8 Hz, 1H), 1.99 (d, J = 1.8 Hz, 3H); IR (KBr) ν: 1739, 1637, 1229, 1267, 1102, 977, 696 cm^{-1}; MS (EI) *m/z* (%) 300 (M^+), 173 (100); Anal. calcd for $C_{11}H_9IO_2$: C 44.03, H 3.02, I 42.29; found C 43.91, H 3.04, I 42.17.

3-(*n*-*Propyl*)-4-*iodo*-5-*phenyl*-2(5*H*)-*furanone* (**2e**) Starting from 1.097 g (5.43 mmol) of **1e**, 1.448 g (5.70 mmol) of iodine, and 0.826 g (6.00 mmol) of K_2CO_3 to afford 1.724 g (97%) of **2e**. White solid, m.p. 104—105℃ (*n*-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 7.48—7.40 (m, 3H), 7.28—7.20 (m, 2H), 5.75 (s, 1H), 2.44 (t, J = 7.3 Hz, 2H), 1.73—1.60 (m, 2H), 1.00 (t, J = 7.3 Hz, 3H); IR (KBr) ν: 1744, 1577, 1246, 1172, 1003 cm^{-1}; MS (EI) *m/z* (%) 328 (M^+), 201 (100); Anal. calcd for $C_{13}H_{13}IO_2$: C 47.58, H 3.99; found C 47.51, H 3.91.

3-*Methyl*-4-*iodo*-5-(1′-*naphthyl*)-2(5*H*)-*furanone* (**2f**) Starting from 448 mg (2.0 mmol) of **1f**, 610 mg (2.4 mmol) of iodine, and 276 mg (2.0 mmol) of K_2CO_3 to afford 682 mg (97%) of **2f**. Yellow solid, m.p. 158.5—159.5℃ (*n*-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 8.10 (d, J = 7.9 Hz, 1H), 7.91 (d, J = 8.0 Hz, 2H), 7.68—7.50 (m, 3H), 7.36 (d, J = 8.2 Hz, 1H), 6.17 (s, 1H), 2.09 (s, 3H); IR (KBr) ν: 1739, 1637, 977, 696 cm^{-1}; MS (EI) *m/z* (%): 350 (M^+), 349 (100); Anal. calcd for $C_{11}H_{17}IO_2$: C 51.45, H 3.17; found C 51.46, H 3.10.

4-*Iodo*-5-*methyl*-5-*phenyl*-2(5*H*)-*furanone* (**2g**) Starting from 348.0 mg (2.0 mmol) of **1g**, 533.4 mg (2.1 mmol) of iodine, and 276.4 mg (2.0 mmol) of K_2CO_3 to afford 522.0 mg (87%) of **2g**. White solid, m.p. 103—104℃ (*n*-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 7.48—7.40 (m, 5H), 6.48 (s, 1H), 1.93 (s, 3H); IR (KBr) ν: 1739, 1646, 1167, 977 cm^{-1}, MS (EI) *m/z* (%): 300 (M^+, 44.01), 173 (100); Anal. calcd for $C_{11}H_9IO_2$: C 44.03, H 3.02, I 42.29; found C 44.17, H, 3.23, I, 42.36.

3-(*n*-*Propyl*)-4-*iodo*-5-(4′-*bromophenyl*)-2(5*H*)-*furanone* (**2h**) Starting from 2.30 g (8.2 mmol) of **1h**, 2.50 g (9.8 mmol) of iodine, and 1.36 g (9.8 mmol) of K_2CO_3 to afford 3.21 g (96%) of **2h**. White solid, m.p. 127.5—128.5℃ (*n*-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 7.53 (d, J = 8.4 Hz, 2H),

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7.10 (d, J = 8.4 Hz, 2H), 5.69 (s, 1H), 2.38 (t, J = 7.5 Hz, 2H), 1.70—1.56 (m, 2H), 0.98 (t, J = 7.5 Hz, 3H); IR (KBr) ν: 1746, 1633, 1490, 1312, 1115, 993, 818; MS (EI) m/z (%): 406 [M^+ (^{79}Br), 17.21], 408 [M^+ (^{81}Br), 16.71], 279 (100), 281 (98.13); Anal. calcd for $C_{13}H_{12}BrIO_2$: C 38.36, H 2.97, Br 19.63, I 31.18; found C 38.35, H 2.94, Br 19.66, I 30.99.

General procedure for the preparation of 3-*methyl*-4-*bromo*-5-*phenyl*-2(5*H*)-*furanone* (**3c**)

A solution of 2-methyl-4-phenyl-2, 3-butadienoic acid (**1d**) (174.0 mg, 1.0 mmol), NBS (214.0 mg, 1.2 mmol) and K_2CO_3 (138.1 mg, 0.5 mmol) in 5 mL of H_2O was stirred at rt for 1 h. The reaction mixture was extracted with diethyl ether (3 × 10 mL), and dried over anhydrous $MgSO_4$. After removal of the solvent, the residue was purified by flash chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 10/1) to afford 237.5 mg (94%) of 3-methyl-4-bromo-5-phenyl-2(5*H*)-furanone (**3c**). Yellow solid, m.p. 137.5—138.5℃ (n-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 7.44—7.38 (m, 3 H), 7.30—7.20 (m, 2H), 5.78 (q, J = 1.7 Hz, 1H), 1.98 (d, J = 2.1 Hz, 3H); IR (KBr) ν: 1760, 1610, 1120 cm^{-1}; MS (EI) m/z (%): 254 [M^+ (^{81}Br)], 252 [M^+ (^{79}Br)], 173 (100); Anal. calcd for $C_{11}H_{19}BrO_2$: C 52.20, H 3.58, Br 31.57; found C 52.11, H 3.56, Br 31.25.

The data for compounds **3a** and **3b** were reported in Ref. 7.

3-(*n*-*Propyl*)-4-*bromo*-5-*phenyl*-2(5*H*)-*furanone* (**3d**) Starting from 202.0 mg (1.0 mmol) of **1e**, 195.8 mg (1.1 mmol) of NBS, and 69.0 mg (0.5 mmol) of K_2CO_3 to afford 277.0 mg (99%) of **3d**. White solid. m.p.: 80—81℃ (n-hexane); ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.44—7.41 (m, 3H), 7.26—7.25 (m, 2H), 5.76 (s, 1H), 2.43 (t, J = 7.5 Hz, 2H), 1.66 (m, 2H), 0.99 (t, J = 7.4 Hz, 3H); MS (EI) m/z (%): 280 [M^+ (^{79}Br), 3.32], 282 [M^+ (^{81}Br), 2.92], 128(100); IR (neat) ν: 1747, 1648, 1456, 1290, 1102, 1067, 995, 766, 697 cm^{-1}; Anal. calcd for $C_{13}H_{13}BrO_2$: C 55.54, H 4.66, Br 28.42; found C 55.47, H 4.66, Br 28.30.

3-*Methyl*-4-*bromo*-5-(1′-*naphthyl*)-2(5*H*)-*furanone* (**3e**) Starting from 224.0 mg (1.0 mmol) of **1f**, 214.0 mg (1.2 mmol) of NBS, and 138.2 mg (1.0 mmol) of K_2CO_3 to afford 297.3 mg (98%) of **3e**. White solid, m.p. 141—141.5℃ (n-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 8.12 (d, J = 8.2 Hz, 1H), 7.93 (d, J = 8.0 Hz, 2H), 7.64—7.46 (m, 3H), 7.35 (d, J = 7.3 Hz, 1H), 6.66 (q, J = 1.7 Hz, 1H), 2.09 (d, J = 1.7 Hz, 3H); IR (KBr) ν: 1760, 1610, 1120 cm^{-1}; MS (EI), m/z (%): 301 [M^+ (^{79}Br), 91.19], 303 [M^+ (^{81}Br), 100]; Anal. calcd for $C_{11}H_{17}O_2$: C 59.43, H 3.66; found C 59.40, H 3.59.

4-*Bromo*-5-*methyl*-5-*phenyl*-2(5*H*)-*furanone* (**3f**)[11] Starting from 174.0 mg (1.0 mmol) of **1g**, 195.6 mg (1.1 mmol) of NBS, and 69.0 mg (0.5 mmol) of K_2CO_3 to afford 252.4 mg (99%). White solid, m.p. 137.5—138.5℃ (n-hexane); ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 7.42—7.34 (m, 5H), 6.29 (s, 1H), 1.96 (s, 3H); MS (EI) m/z (%): 252 [M^+ (^{79}Br), 8.13], 254 [M^+ (^{81}Br), 8.66], 173 (100); IR (KBr) ν: 1745, 1601, 1450, 1234, 945, 860, 771 cm^{-1}.

General procedure for the preparation of 4-*aryl*-5-(*n*-*butyl*)-2(5*H*)-*furanone* (**4a**)

To a solution of 2 mL of toluene and 0.1 mL of MeOH was added phenylboronic acid (60.0 mg, 0.50 mmol) and 4-iodo-5-(*n*-butyl)-2(5*H*)-furanone (**2a**) (66.5 mg, 0.25 mmol). Then 0.3 mL of aqueous sodium carbonate solution (2 mol/L in H_2O) was added. The reaction system was degassed and refluxed overnight. The reaction was monitored by TLC until the reaction was over. The reaction mixture was quenched with water and extracted with diethyl ether (3 × 15 mL). The organic layers were combined and washed with brine and dried over anhydrous $MgSO_4$. After removal of the solvent, the residue was purified by flash chromatography (eluent: petroleum ether/ethyl acetate = 10/1) to afford 50.8 mg (93%) of 4-aryl-5-(*n*-butyl)-2(5*H*)-furanone (**4a**).[2,12] Yellow oil, ^{1}H NMR (300 MHz, $CDCl_3$) δ: 7.42—7.48 (m, 5H), 6.25 (d, J = 1.4 Hz, 1H), 5.50 (ddd, J = 7.8, 2.8 and 1.4 Hz, 1H), 1.95—2.05 (m, 1H), 1.50—1.60 (m, 1H), 1.20—1.46 (m, 4H), 0.85 (t, J = 7.2 Hz, 3H).

4-(3′-*Methoxylphenyl*)-5-(*n*-*butyl*)-2-(5*H*)-*furanone* (**4b**) Starting from 66.5 mg (0.25 mmol) of **2a** and 75.0 mg (0.5 mmol) of *o*-methoxylphenylboronic acid to afford 54.0 mg (88%) of **4b**. Yellow oil; ^{1}H NMR ($CDCl_3$, 300 MHz) δ: 7.40 (t, J = 8.0

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Hz, 1H), 7.02—6.94 (m, 3H), 6.24 (s, 1H), 5.50—5.40 (m, 1H), 3.88 (s, 3H), 2.04—1.96 (m, 1H), 1.62—1.53 (m, 1H), 1.48—1.24 (m, 4H), 0.84 (t, J = 7.0 Hz, 3H); IR (neat) ν: 1748, 1634, 1466, 1173 cm^{-1}; MS (EI) m/z (%): 246 (M^+, 62.60), 133 (100); HRMS calcd for $C_{15}H_{18}O_3$: 246.1256; found 246.1256.

4-Phenyl-5-(n-heptyl)-2-(5H)-furanone (**4c**)[2] Starting from 77.0 mg (0.25 mmol) of **2b** and 60.0 mg (0.5 mmol) of phenylboronic acid to afford a yellow oil (56.8 mg, 88%) of **4c**. ^1H NMR ($CDCl_3$, 300 MHz) δ: 7.40—7.50 (m, 5H), 6.28 (d, J = 1.4 Hz, 1H), 5.49 (ddd, J = 7.6, 2.7 and 1.4 Hz, 1H), 1.95—2.03 (m, 1H), 1.46—1.62 (m, 1H), 1.10—1.35 (m, 10 H), 0.82 (t, J = 7.2 Hz, 3H).

3-Methyl-4,5-diphenyl-2(5H)-furanone (**4d**)[12] Starting from 78.0 mg (0.25 mmol) of **2d** and 60.0 mg (0.5 mmol) of phenylboronic acid to afford a yellow oil 61.9 mg (99%) of **4d**; Starting from 68.7 mg (0.25 mmol) of **3c** and 60.0 mg (0.5 mmol) of phenyl boronic acid to afford 60.6 mg (97%) of **4d**. Yellow oil. ^1H NMR (300 MHz, $CDCl_3$) δ: 7.12—7.38 (m, 10H), 6.19 (q, J = 1.6 Hz, 1H), 2.17 (d, J = 1.6 Hz, 3H); IR (neat) ν: 1746, 1648, 1010 cm^{-1}; MS (EI) m/z (%): 250 (M^+, 42.33), 145 (100).

3-Methyl-4-(p-methoxylphenyl)-5-phenyl-2(5H)-furanone (**4e**) Starting from 60.0 mg (0.20 mmol) of **2d** and 60.0 mg (0.4 mmol) of *p*-methoxylphenylboronic acid to afford 46.7 mg (85%) of **4e**. Yellow oil. ^1H NMR ($CDCl_3$, 300 MHz) δ: 7.38—7.20 (m, 7H), 6.90 (d, J = 8.9 Hz, 2H), 6.19 (q, J = 1.5 Hz, 1H), 3.83 (s, 3H), 2.20 (d, J = 1.6 Hz, 3H); IR (neat) ν: 1746, 1606, 1258, 1038 cm^{-1}; MS (EI) m/z (%): 280 (M^+, 22.04), 69 (100); HRMS cacld for $C_{18}H_{16}O_3$: 280.1095; found 280.1111.

3-Methyl-4-phenyl-5-(1'-naphthyl)-2(5H)-furanone (**4f**) Starting from 70.0 mg (0.20 mmol) of **2f** and 48.0 mg (0.4 mmol) of phenyl boric acid to afford 52.3 mg (87%) of **4f**; Starting from 60.6 mg (0.20 mmol) of **3e** and 48.0 mg (0.4 mmol) of phenylboronic acid to afford 49.8 mg (83%) of **4f**. Yellow oil; ^1H NMR ($CDCl_3$, 300 MHz) δ: 8.28 (d, J = 8.4 Hz, 1H), 7.88 (d, J = 7.8 Hz, 1H), 7.80 (d, J = 8.3 Hz, 1H), 7.68—7.52 (m, 2H), 7.38—7.25 (m, 6H), 7.12 (dd, J = 7.1 Hz and 0.9 Hz, 1H), 7.05 (d, J = 1.6 Hz, 1H) 2.29 (d, J = 1.6 Hz, 3H); MS (EI) m/z (%): 300 (M^+, 62.12), 279 (100); IR (neat) ν: 1747, 1651, 1444, 1259, 1100, 1035, 976, 914, 782, 766, 736, 699 cm^{-1}; HRMS cacld for $C_{21}H_{16}O_2$: 300.11503; found 300.11562.

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